

## DeepHalo Documentation

**DeepHalo: A deep learning-integrated workflow for high-throughput discovery of halogenated metabolites from HRMS data.**

*Version 1.0.1*

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### Introduction

DeepHalo is a hierarchically optimized workflow designed for the detection and dereplication of halogenated compounds in **high-resolution** mass spectrometry (HRMS) data. It integrates cutting-edge deep learning models, robust isotope validation, comprehensive scoring, and dual dereplication strategies to deliver high accuracy and efficiency. Key applications span multiple fields, ranging from natural product discovery and halogenated metabolite annotation to precise tracking of halogenated drug metabolism in pharmacological studies; from comprehensive monitoring of persistent halogenated pollutants in environmental science to sensitive detection of chlorinated disinfection byproducts in water quality analysis.

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### Data Support

DeepHalo supports **high-resolution** mass spectrometry (HRMS) data, with or without tandem mass spectrometry (MS/MS) information. Note that the MS<sup>2</sup> extraction function currently supports only data-dependent acquisition data (DDA). Additionally, DeepHalo

exclusively supports the **.mzML** file format for analysis.

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## Core Features

### 1. Halogen Prediction

- **Element Prediction Model (EPM)**
  - Dual-branch Isotope Neural Network (IsoNN) architecture
  - High accuracy Cl/Br detection (>99.9% precision based on benchmark results)
  - Wide mass range coverage (50-2000 Da)
  - Robust interference resistance to B/Se/Fe/dehydro isomers

### 2. Isotope Pattern Validation

- **Dual Validation System**
  - *Mass Dimension*: Statistical rule-based correction.
  - *Intensity Dimension*: Autoencoder-based Anomaly Detection Model (ADM).

### 3. Multi-Level Halogen Confidence Scoring (H-score)

- **- Dual levels**
  - Prediction based on centroid-level isotope patterns
  - Prediction based on Scan-level isotope patterns
  - H-score integration for comprehensive assessment on the above both levels

### 4. Dereplication

- **Dual Strategy Approach**
    - *MS1-based dereplication using Custom Database Matching*: Validates exact mass, halogen patterns, and isotope intensity similarity.
    - *MS2-Based Dereplication by Integrating GNPS*: MS2 molecular networking, halogenated compound annotation, graphML file enhancement.
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## Technical Advantages

- **High Throughput**: automatically process unlimited samples in several to dozens of seconds each on standard hardware (Core i9, 16GB RAM).
  - **Accuracy**: >98.3% precision in halogen detection across simulated and experimental LC-MS datasets.
  - **Integration with GNPS**: enhance molecular network annotation in the element dimension by embedding DeepHalo results into GNPS output graphML file
  - **Efficient Dereplication**: significantly higher dereplication rate compared to molecular networking alone in GNPS
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## Installation

### For EXE Users:

- Double click the exe to launch DeepHalo.
- Available at [https://github.com/xieyying/deephalo/releases/tag/DeepHalo\\_V1.0.0](https://github.com/xieyying/deephalo/releases/tag/DeepHalo_V1.0.0))  
and <https://pan.baidu.com/s/1Bxr1aEGl7GDZ74XHBdA1IA?pwd=tyg1>

### For WHL Users:

#### Prerequisites

- Python 3.10 (Verify with `python --version`).

#### Installation Methods

##### From PyPI

```
pip install deephalo
```

##### From Local Wheel

```
pip install path/to/DeepHalo-xxx.whl
```

##### From Source

```
git clone https://github.com/xieyying/DeepHalo.git  
cd DeepHalo  
pip install -e .
```

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## Quickstart

### For EXE Users:

Double click the exe to launch DeepHalo.

#### 1. Detect Halogenated Compounds in mzML Files

DeepHalo V1.0.0

**DeepHALO**

A deep learning-integrated workflow for high-throughput discovery of halogenated metabolites from HRMS data

(Note: HRMS data and mzML format needed)

**Detect** Dereplicate

Input (.mzML file or folder):

Output Directory:

Config File (optional):

Blank Folder (optional):

☐ Overwrite Blank

☐ Enable MS2 Extraction (only DDA data supported)

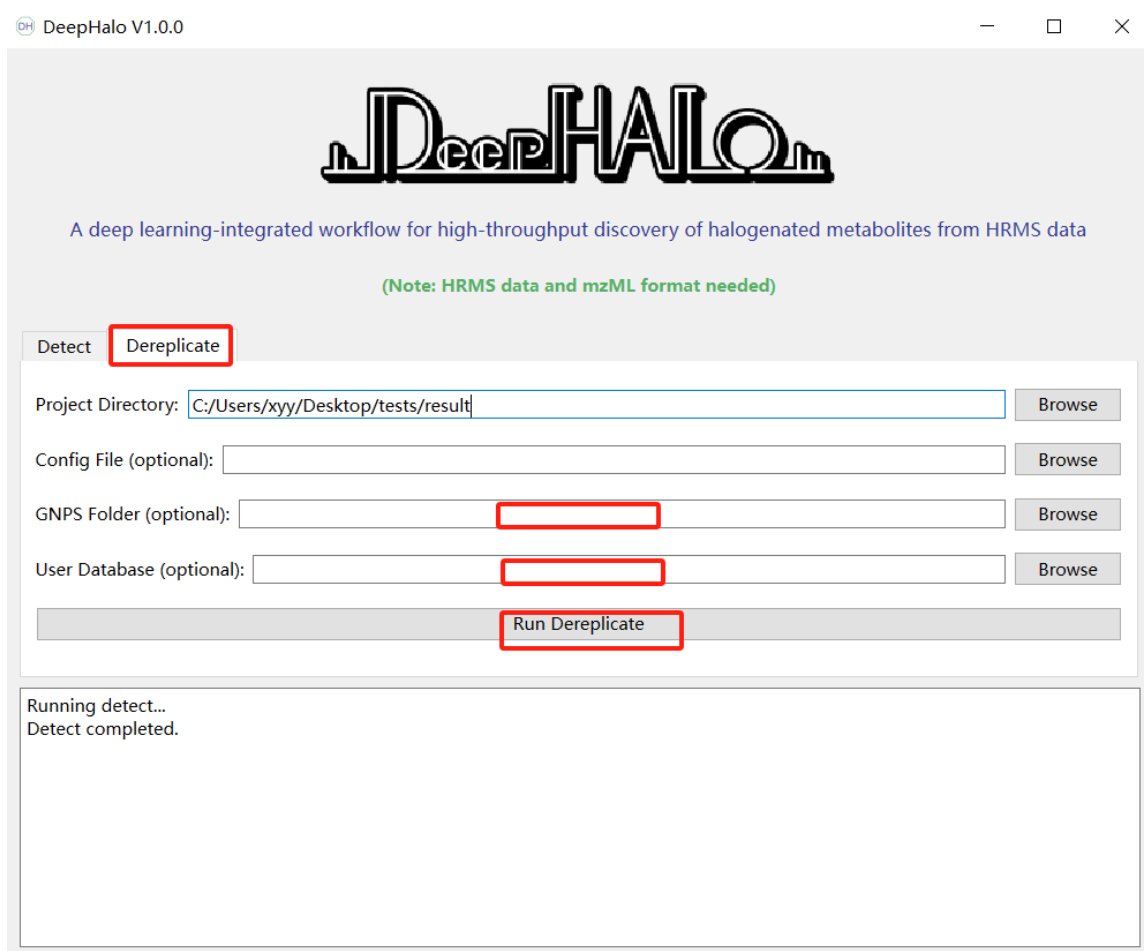
**Note:** Please use English characters for file and directory names

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## 2. Dereplication with GNPS and/or User Database

For dereplication, you must provide at least one of GNPS results or a User Database.”

(Note: User Database format see below Example DATABASE\_FILE)



### For WHL Users:

#### 1. Detect Halogenated Compounds in mzML Files

```
halo detect -i /path/to/mzml_files -o /path/to/output_directory -ms2
```

#### 2. Dereplication with GNPS and/or Custom Database

```
halo dereplicate -o /path/to/output_directory -g /path/to/GNPS_results -ud  
/path/to/custom_database.csv
```

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## Command-Line Usage

### General Help

```
halo --help # List all commands
```

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halo [command] --help                   # Show options for a specific command (e.g., `halo detect --help`)

## Commands

### 1. Analyze mzML Files

halo detect -i <input\_path> -o <project\_path> [-c <config\_file>] [-b <blank\_samples\_dir>] [-ms2]

### 2. Dereplication

halo dereplicate -o <project\_path> [-g <GNPS\_folder>] [-ud <user\_database.csv>]

### 3. Create Training Dataset

halo create-ds <project\_path> [-c <config\_file>]

### 4. Train Model

halo train <project\_path> [-c <config\_file>] [-m search]

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## Output Directory Structure

/output\_directory

```
├──dereplication
│   ├── Demo_data_1_feature.csv
│   └── Demo_data_2_feature.csv
└──result
    ├── config.toml
    ├── error_files.txt
    ├──halo
    │   ├── Demo_data_1_feature.csv
    │   ├── Demo_data_1_scan.csv
    │   ├── Demo_data_2_feature.csv
    │   └── Demo_data_2_scan.csv
    └──ms2_output
        ├── Demo_data_1.mzML
        └── Demo_data_2.mzML
```

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## Instruction

The whole DeepHalo analysis process including 4 steps

### Step 1: Halogenate Mining

#### Run the command:

```
halo detect -i <input_path> -o <project_path> [-c <config_file>] [-b <blank_samples_dir>] [-ob] [-ms2]
```

#### Parameters:

- **-i INPUT\_PATH (Required)**
  - Input .mzML file or directory
  - Example (Linux): -i /home/data/ms\_files
  - Example (Windows): -i D:\data\ms\_files
- **-o OUTPUT\_DIR (Required)**
  - Output directory path
  - Example (Linux): -o /home/analysis/output
  - Example (Windows): -o D:\analysis\output
- **-c <config\_file> (Optional)**
  - Parameters for LCMS data analysis
  - Example(Linux): -c /home/deephalo/config.toml
  - Example (Windows): -c D:\deephalo\config.toml
- **-b <blank\_samples\_dir> (Optional)**
  - Blank .mzML file or directory
  - Example (Linux): -b /home/analysis/blank
  - Example (Windows): -b D:\analysis\blank
  - Run blank exclusion during analysis
  - Default: disabled
- **-ob (Optional)**
  - Force regenerate blank feature detection results for blank exclusion
  - Default: disabled
- **-ms2 (Optional)**
  - Enable MS2 data extraction
  - Optional parameter (required for steps 2-4)
  - Default: disabled

#### Example Usage:

```
halo analyze-mzml -i D:\data\ms_files -o D:\analysis\output -ms2
```

#### Output Files

- config.toml: Analysis parameters

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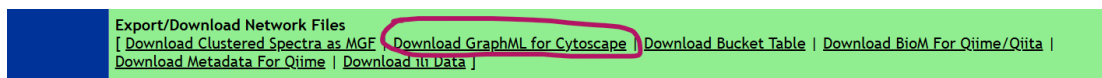
- error\_files.txt: Failed mzML files
- halo\.: MS1 information
- ms2\_output\.: MS2 data

## Notes

- Results are filtered by H-score
- To disable filtering: Set H-score to 0 in config.toml file
- H-score visible in Cytoscape after dereplication
- For failed files:
  1. Retry analysis
  2. If error persists, reconvert raw data to mzML

## Step 2: Molecular Networking analysis employing GNPS

1. Submit MS2 data in ms2\_output to GNPS platform
2. Analyze molecular similarity using GNPS platform
3. Download and unzip results



## Example GNPS output structure:

```
└─ Demo_GNPS_output
  │ METABOLOMICS-SNETS-V2-xxx-main.graphml
  │ params.xml
  └─ clusterinfo
    └─ clusterinfosummarygroup_attributes_withIDs
      └─ gnps_molecular_network_graphml
        └─ result_specnets_DB
```

## Step 3: Dereplication

### Run the command:

```
halo dereplicate -o PROJECT_PATH [-g GNPS_DIR] [-ud DATABASE_FILE]
```

- -o PROJECT\_PATH (**Required**)
  - Output directory (same as used in function of 'detect')
  - Example: -o D:\analysis\output
- -g GNPS\_DIR (**Optional**)
  - GNPS results directory containing .GraphML file
  - Example: -g D:\analysis\Demo\_GNPS\_output



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- -ud DATABASE\_FILE (**Optional**)
  - Custom database file (CSV/JSON)
  - Example: -ud D:\databases\compounds.csv

## Output Details

Upon completion of the analysis, the following files and directories will be generated:

### 1. 'dereplication' Directory

- Contains processed MS1 data in CSV format.
- If a database was provided, compound match information will also be included in the csv file.

### 2. GraphML File (generated when GNPS results are provided)

- The file name will have the suffix \_adding\_DeepHalo\_results.
- This file contains the mean H-score and classification predicted from centroid isotope patterns, along with any database match information if a dereplication database was used.

### 3. Processed Database File (if a database is provided)

- The file name will include the suffix \_DeepHalo\_dereplication\_ready\_database.
- This file is formatted for future analyses, significantly reducing processing time in subsequent runs.

## Example DATABASE\_FILE:

| compound_name | formula | Smiles            |
|---------------|---------|-------------------|
| Compound1     | C6H5Cl  | ClC1=CC=CC=C1     |
| Compound2     | C6H4Cl2 | ClC1=CC=C(Cl)C=C1 |

## Important Notes

### 1. File Format

- The input file must be in CSV format.

### 2. Required Columns

- compound\_name
- formula

### 3. Optional (but Recommended) Column

- Smiles (These aid in structure visualization).

### 4. Additional Requirements

- Ensure that the formula column contains valid molecular formulas.
- The file must be encoded in UTF-8.

## Step 4: Visualization in Cytoscape

To visualize the results, open the GraphML file (with the suffix \*\_adding\_DeepHalo\_results.graphml) in Cytoscape. The file includes the following annotated variables:

- **H\_scoreMean:**
    - **Range:** [0, 1]
    - **Default Threshold:** 0.4, above which the presence of halogen is inferred.
    - **Interpretation:** Higher values indicate a greater likelihood of halogen presence.
    - **Other Factors Affecting the Score:**
      - Sample complexity
      - Mass spectrometer resolution
  - **classification:**
    - **Isotope pattern classifications (0-7):**
      - 0: X-type (mixed/polyhalogenated)
      - 1: Br-type (Br/Cl<sub>3</sub>)
      - 2: Cl-type (Cl/Cl<sub>2</sub>)
      - 3: Se-type (Selenium compounds)
      - 4: B-type (Boron compounds)
      - 5: Fe-type (Iron compounds)
      - 6: C-type (compounds containing only C, H, O, N, F, P, S, Na)
      - 7: artifact-type (artifact isotope patterns)
  - **Inty\_cosine\_score:**

Cosine similarity between experimental and theoretical isotope peak intensities. Higher values indicate a greater likelihood of a known compound
  - **Compound\_names and Adducts:**

These fields provide the known molecular names and adduct information sourced from a user-provided database, offering essential metadata for compound identification.
  - **Smiles**

Provides the known molecular structure, sourced from either the GNPS library and a user-provided database.
  - **error\_ppm**

Represents the mass error between the measured m/z value of the test molecule and that of its corresponding known compound match.
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Table 1: Key Data Processing Parameters in DeepHalo

| Module / Section                               | Parameter                   | Value / Setting (default) | Description                                                                                           |
|------------------------------------------------|-----------------------------|---------------------------|-------------------------------------------------------------------------------------------------------|
| <b>FeatureFilter</b>                           | RT_min                      | "                         | Minimum retention time (in seconds) to keep MS data. Leave empty (") to keep all data (no RT filter). |
|                                                | RT_max                      | "                         | Maximum retention time (in seconds) to keep MS data. Leave empty (") to keep all data (no RT filter). |
|                                                | H_score_threshold           | 0.4                       | Confidence threshold for halogenated compound detection (Range: 0-1, higher is stricter).             |
|                                                | Anomaly_detection_threshold | 7.02e-05                  | Threshold for isotope peak intensity reconstruction error. Use 1.0 to disable filtering.              |
| <b>FeatureMapProcessor</b>                     | min_num_of_masstraces       | 3                         | Minimum number of isotope peaks a feature must have.                                                  |
|                                                | min_feature_int             | 5000                      | Minimum summed intensity of a candidate feature.                                                      |
|                                                | min_scan_number             | 3                         | Minimum number of consecutive scans a feature must span.                                              |
|                                                | use_mass_difference         | " (empty)                 | Enables mass difference correction for isotope patterns. Set to 'none' to disable.                    |
| <b>Dereplication</b>                           | error_ppm                   | 10                        | Precursor mass tolerance (ppm) for database matching.                                                 |
|                                                | Inty_cosine_score           | 0.96                      | Minimum cosine similarity threshold between experimental and theoretical isotope intensities.         |
| <b>FeatureFinding (Mass Trace Detection)</b>   | mass_error_ppm              | 10.0                      | Mass error (ppm) for grouping ions into mass traces.                                                  |
|                                                | min_trace_length            | 3.0                       | Minimum number of data points required for a mass trace.                                              |
|                                                | noise_threshold_int         | 1000.0                    | Intensity threshold below which signals are considered noise.                                         |
|                                                | chrom_peak_snr              | 6.0                       | Signal-to-noise ratio threshold for chromatographic peak detection.                                   |
| <b>FeatureFinding (Elution Peak Detection)</b> | chrom_fwhm                  | 3.0                       | Expected chromatographic peak width (full width at half maximum).                                     |
|                                                | min_fwhm / max_fwhm         | 1.0 / 60.0                | Minimum and maximum allowed peak width.                                                               |
| <b>FeatureFinding (Feature Detection)</b>      | local_mz_range              | 5.5                       | m/z range for grouping mass traces into features.                                                     |
|                                                | mz_scoring_by_elements      | true                      | Enables scoring by elemental composition (essential for DeepHalo).                                    |
|                                                | elements                    | "CHNOPSFCIBrINaFeSeB"     | Elements considered for theoretical isotope pattern generation.                                       |
|                                                | remove_single_traces        | true                      | Removes features consisting of only a single mass trace (essential for DeepHalo).                     |
|                                                | isotope_filtering_model     | "none"                    | Disables default isotope filtering, as DeepHalo performs its own classification.                      |

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## Dependencies,

- pandas == 2.0.3
  - numpy == 1.22.0
  - tensorflow == 2.10.1
  - scikit-learn == 1.3.1
  - pyopenms == 3.1.0
  - Full list: See [README.md](#)
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## License

Distributed under the [MIT License](#).

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*For methodology details and benchmarks, refer to the [GitHub repository](#).*